SIMATS ENGINEERING

ASSESSMENT TOOL 2

Case study

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Answers

1 **Sol:**

Analyze whether the problem is caused by the Transport Protocol or the Application Layer.

The problem is primarily caused by the Transport Protocol (UDP) and is further affected by the Application Layer's buffering strategy.

- Real-time video conferencing applications (Zoom, Google Meet, Microsoft Teams) use UDP because it provides low latency and does not retransmit lost packets.
- An 8% packet loss and 180 ms jitter directly affect UDP communication, causing missing audio/video packets.
- If the application's jitter buffer is too small, delayed packets are discarded, resulting in voice distortion and frozen video.

Therefore, the issue involves both the transport layer (UDP) and the application layer (buffer management).

2. Justification

Real-time multimedia applications require:

- Low latency
- Continuous media streaming
- Minimal delay

Why UDP is used

- No connection establishment
- No retransmissions
- Lower overhead
- Faster packet delivery

If TCP were used:

- Lost packets would be retransmitted.
- Video and audio would be delayed.
- User experience would become worse.

Effect of Network Issues

Problem	**Impact**
8% Packet Loss	Missing audio/video packets
180 ms Jitter	Packets arrive at irregular intervals
Bandwidth Fluctuation	Video quality automatically decreases
Small Jitter Buffer	Voice breaks and frozen frames

Hence, UDP is suitable, but poor network quality significantly affects conferencing performance.

3. Recommended Improvements

Transport Layer Improvements

- Enable QoS (Quality of Service) to prioritize voice and video packets.
- Use RTP over UDP with packet loss recovery techniques.
- Implement Forward Error Correction (FEC).

- Apply adaptive bitrate streaming.

Application Layer Improvements

- Increase adaptive jitter buffer size.
- Use video compression (H.265/VP9).
- Dynamically adjust video resolution.
- Enable packet loss concealment.

Expected Result

- Reduced latency
- Smooth audio
- Better video quality
- Improved user experience

2 Sol:

DNS Performance Optimization

Scenario

Average DNS lookup time = 800 ms
Target = Less than 50 ms

1. Reasons for High DNS Resolution Time

Possible causes include:

- Single DNS server handling all requests
- High network latency
- No DNS caching
- Large geographic distance between clients and DNS server
- Incorrect TTL configuration
- DNS server overload

These factors increase response time and delay webpage loading.

2. Proposed DNS Architecture

a) Anycast DNS

Deploy multiple DNS servers globally.
Users are automatically routed to the nearest DNS server.
Benefits:

- Faster responses
- Lower latency
- Load balancing
- High availability

b) DNS Caching

Store previously resolved domain names locally.
Benefits:

- Eliminates repeated DNS lookups
- Faster website access
- Reduces DNS server load

c) DNS Pre-fetching

Browsers resolve domain names before users click links.
Benefits:

- Immediate connection establishment
- Reduced waiting time

d) TTL Optimization

Set appropriate TTL values.
Example:

- Static websites → High TTL
- Dynamic services → Medium TTL

Benefits:

- Faster cached responses
- Reduced DNS traffic

Expected Lookup Time

Current = 800 ms

After optimization = 20–50 ms

3. Advantages and Limitations

Advantages

- Faster DNS resolution
- Improved scalability
- Load balancing
- Better fault tolerance
- Higher availability
- Reduced server load

Limitations

- Higher deployment cost
- More complex management
- Incorrect TTL values may cause stale records
- Requires global infrastructure

3 **Sol:**

Distributed DNS for Cloud Services Scenario

All DNS requests are handled by a single primary DNS server.

Average lookup time = 800 ms

1. Impact of Centralized DNS

Centralized DNS creates:

- High latency
- Server overload
- Single point of failure
- Slow application loading
- Reduced scalability

When thousands of users query simultaneously, the DNS server becomes a bottleneck.

2. Distributed DNS Solution

Recommended Architecture:

```
         Users
           |
     Anycast Routing
  ------------------------
  |        |         |
DNS US   DNS Europe   DNS Asia
    \      |      /
      Cloud Services
```

Features:

- Multiple geographically distributed DNS servers
- Anycast routing
- Local DNS caching
- Secondary DNS servers

- Automatic failover

Benefits:

- Low latency
- High availability
- Better scalability
- Improved fault tolerance

3.Effect of TTL, Anycast and DNS Caching

TTL

Low TTL → Faster updates but more DNS queries

Anycast Routing

Routes users to the nearest DNS server.

Benefits:

- Lower latency
- Automatic load balancing
- Fast failover

DNS Caching

Stores previous DNS responses.

Benefits:

- Faster lookups
- Reduced DNS traffic
- Lower server load

Overall Performance

- Faster application loading
- Better reliability
- Improved fault tolerance
- High availability

3. Stock Trading Platform – TCP Latency Analysis

Scenario

Order acknowledgment latency increases from 1 ms to 40 ms.

Observed issues:

- TCP retransmissions
- Slow connection establishment
- Excessive buffering

4 Sol:

Three TCP-Related Factors

(a) Flow Control

TCP adjusts the transmission rate based on the receiver's buffer size.

Effect:

- Sender slows down when the receiver is overloaded.
- Orders are delayed.

(b) Nagle's Algorithm

Nagle combines multiple small packets into a larger packet before sending.

Effect:

- Reduces network overhead.
- Introduces delays for small, frequent trading messages.

(c) SYN Queue Limitation

During heavy traffic, many TCP connection requests arrive simultaneously.

If the SYN queue becomes full:

- New connections are dropped.
- Clients retry.
- Connection establishment is delayed.

2. Effect on High-Frequency Trading

TCP Factor	Impact
Flow Control	Slows transmission when buffers are full
Nagle's Algorithm	Adds delay by waiting to combine packets
SYN Queue Overflow	Delays or rejects new TCP connections

These issues increase transaction latency and reduce trading performance.

3. Recommended TCP Configuration Changes

Disable Nagle's Algorithm

Enable TCP_NODELAY to send packets immediately.

Increase SYN Queue Size

Allows more simultaneous connection requests during peak trading.

Optimize TCP Buffers

Increase send and receive buffer sizes to reduce congestion.

Enable TCP Fast Open

Reduces connection setup time by allowing data transfer during the initial handshake.

Use Persistent Connections

Reuse existing TCP connections instead of creating a new connection for every order.

Enable QoS

Prioritize trading traffic over less critical network traffic.

Expected Benefits

- Lower acknowledgment latency
- Faster order processing
- Fewer retransmissions
- Improved transaction reliability
- Better customer experience